

Exercise 1.

Show that axiom (M0) in the definition of a metric space is not needed, i.e. assuming that axioms (M1) to (M4) hold, prove that (M0) has to hold.

Solution. Putting $z = x$ into (M4) we obtain that $d(x, x) \leq d(x, y) + d(y, x)$, for all $x, y \in X$. Using (M2) and (M3) implies that $0 \leq 2d(x, y) \implies d(x, y) \geq 0$, for all $x, y \in X$. ■

Exercise 2.

Prove the triangle inequality (M4) for Euclidean n -space:

$$d_2(x, z) \leq d_2(x, y) + d_2(y, z), \quad \forall x, y, z \in \mathbf{R}^n,$$

by first letting $r_i := x_i - y_i$ and $s_i := y_i - z_i$. So we need to prove

$$\left(\sum_{i=1}^n (r_i + s_i)^2 \right)^{\frac{1}{2}} \leq \left(\sum_{i=1}^n r_i^2 \right)^{\frac{1}{2}} + \left(\sum_{i=1}^n s_i^2 \right)^{\frac{1}{2}}.$$

First show that this is equivalent to **Cauchy's Inequality**:

$$\left(\sum_{i=1}^n r_i s_i \right)^2 \leq \sum_{i=1}^n r_i^2 \sum_{i=1}^n s_i^2.$$

Now prove **Cauchy's Inequality**. Hint: You may want to consider that, for all $\lambda \in \mathbf{R}$, $\sum_{i=1}^n (r_i + \lambda s_i)^2 \geq 0$.

Solution. To obtain Cauchy's Inequality, square both sides of the equation to get

$$\sum_{i=1}^n (r_i + s_i)^2 = \sum_{i=1}^n r_i^2 + 2 \sum_{i=1}^n r_i s_i + \sum_{i=1}^n s_i^2 \leq \sum_{i=1}^n r_i^2 + 2 \left(\sum_{i=1}^n r_i^2 \right)^{\frac{1}{2}} \left(\sum_{i=1}^n s_i^2 \right)^{\frac{1}{2}} + \sum_{i=1}^n s_i^2,$$

which yields the result. To prove Cauchy's Inequality notice that, by the hint,

$$f(\lambda) := \lambda^2 \sum_{i=1}^n s_i^2 + 2\lambda \sum_{i=1}^n r_i s_i + \sum_{i=1}^n r_i^2 \geq 0.$$

Which implies that the quadratic equation $f(\lambda) = 0$ cannot have two distinct real roots. So the “discriminant” $b^2 - 4ac$ must be non-positive which gives the result.

For an alternative proof of Cauchy's Inequality, one can use the Cauchy–Schwarz inequality for the standard dot product on $\mathbf{R}^n$, see MTH107 Advanced Linear Algebra. ■

Exercise 3.

Show that the following function defines a metric on $\mathbf{R}^n$:

$$d_\infty(x, y) := \max\{|x_1 - y_1|, \dots, |x_n - y_n|\}.$$

Solution. The axioms (Mo) obviously holds. For (M1) and (M2), $d_\infty(x, y) = 0$ if and only if $|x_i - y_i| = 0$ for all $1 \leq i \leq n$, if and only if $x = y \in \mathbf{R}^n$. (M3) follows from $|x - y| = |y - x|$. For (M4), for each $1 \leq i \leq n$,

$$|x_i - z_i| \leq |x_i - y_i| + |y_i - z_i| \leq d_\infty(x, y) + d_\infty(y, z),$$

so $d_\infty(x, z) \leq d_\infty(x, y) + d_\infty(y, z)$. ■

Exercise 4.

Determine whether or not the following functions d define metrics on the indicated sets:

- a. $d(x, y) = |x^2 - y^2|$, for $x, y \in \mathbf{R}$;
- b. $d(x, y) = |x^2 - y^2|$, for $x, y \in [0, \infty) \subseteq \mathbf{R}$;
- c. $d(x, y) = (x_1 - y_1)^2 + (x_2 - y_2)^2$, for $x, y \in \mathbf{R}^2$.

Solution. a. No, since $d(x, -x) = 0$ for $x \neq 0$ which goes against (M2).

b. Yes. For the triangle inequality $|x^2 - z^2| \leq |x^2 - y^2| + |y^2 - z^2|$ for all $x, y, z \geq 0$.

c. No. Notice that

$$4 = d((2, 0), (0, 0)) > d((2, 0), (1, 0)) + d((1, 0), (0, 0)) = 1 + 1 = 2;$$

so the triangle inequality fails. ■

Exercise 5.

Show that the British Railway^a metric on $\mathbf{R}^2$, given by

$$d(x, y) := \begin{cases} d_2(x, y), & \text{if } x \text{ and } y \text{ are linearly dependent;} \\ d_2(x, 0) + d_2(0, y), & \text{otherwise,} \end{cases}$$

satisfies the axioms of a metric space, where d_2 is the Euclidean metric on $\mathbf{R}^2$.

^aAlso known as the French Railway metric.

Solution. The axioms (Mo)–(M3) are easy to check. For (M4), first notice that $d_2(x, y) \leq d_2(x, 0) + d_2(0, y)$, for all $x, y \in \mathbf{R}^2$. If x and z are linearly dependent then

$$d(x, z) = d_2(x, z) \leq d_2(x, y) + d_2(y, z) \leq d(x, y) + d(y, z).$$

If x and z are linearly independent, then either $\{x, y\}$ and $\{y, z\}$ are both linearly independent sets or y is dependent on either x or z . In the first case, (M4) is equivalent to $d_2(0, y) \geq 0$, which always holds, and in the second case (if we assume that x and y are dependent) then

$$d(x, z) = d_2(x, 0) + d_2(0, z) \leq d_2(x, y) + d_2(y, 0) + d_2(0, z) = d(x, y) + d(y, z). \quad \blacksquare$$

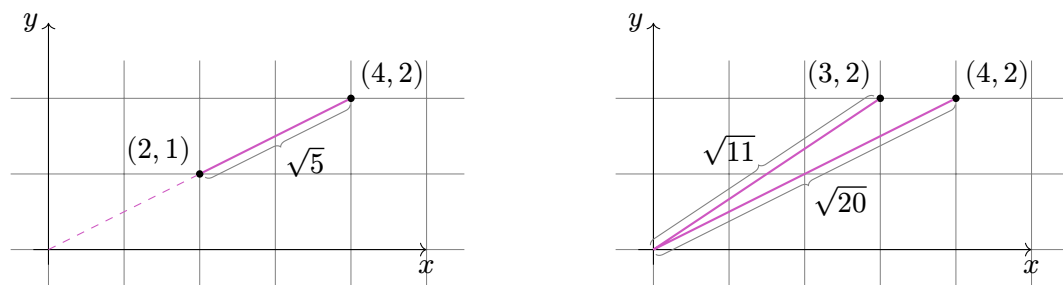


Figure 1: British railway distance between two points in the plane. We have $d((2, 1), (4, 2)) = \sqrt{5}$ while $d((3, 1), (4, 2)) = \sqrt{11} + \sqrt{20}$.

Exercise 6.

Given three points $x, y, z \in X$ in a metric space (X, d) , show that

$$|d(x, z) - d(y, z)| \leq d(x, y).$$

Solution. It's equivalent to $-d(x, y) \leq d(x, z) - d(y, z) \leq d(x, y)$ and each inequality is equivalent to the triangle inequality. ■

Exercise 7.

Show that the following functions between metric spaces are continuous.

- The identity function: $\text{Id}_X: (X, d) \rightarrow (X, d)$, where $\text{Id}_X(x) := x$, for all $x \in X$.
- Inclusion: $i: (W, d|_W) \rightarrow (X, d)$, where $W \subseteq X$ and $i(w) := w$, for all $w \in W$.
- Constant function: $f: (X, d_X) \rightarrow (Y, d_Y)$, where $f(x) := y_0$, for all $x \in X$ and some fixed $y_0 \in Y$.
- Show that every isometry is continuous.

Solution. a. Take $\delta = \varepsilon$.

b. Take $\delta = \varepsilon$.

c. Take any $\delta > 0$.

d. Since $d_Y(f(x), f(a)) = d_X(x, a)$ we can take $\delta = \varepsilon$. ■

Exercise 8.

Suppose that $f: (X, d_X) \rightarrow (Y, d_Y)$ and $g: (Y, d_Y) \rightarrow (Z, d_Z)$ are continuous functions. Show that the composition

$$g \circ f: (X, d_X) \rightarrow (Z, d_Z),$$

where $(g \circ f)(x) := g(f(x))$, for all $x \in X$, is continuous.

Solution. Let $\varepsilon > 0$ and $a \in X$. We need to find $\delta > 0$ such that

$$d_X(x, a) < \delta \implies d_Z(g(f(x)), g(f(a))) < \varepsilon.$$

Since g is continuous at $f(a) \in Y$ there exists $\delta_g > 0$ such that

$$d_Y(y, f(a)) < \delta_g \implies d_Z(g(y), g(f(a))) < \varepsilon.$$

Also, f is continuous at $a \in X$ so there exists $\delta_f > 0$ such that

$$d_X(x, a) < \delta_f \implies d_Y(f(x), f(a)) < \delta_g.$$

We can then take $\delta = \delta_f$ to get what we need by setting $y = f(x)$. ■

Exercise 9.

Let (X, d_X) be a metric space and fix $x_0 \in X$. Show that the function

$$f: (X, d_X) \implies (\mathbf{R}, d_{\mathbf{R}}),$$

given by $f(x) := d_X(x, x_0)$, for all $x \in X$, is continuous (where $d_{\mathbf{R}}$ is the usual metric on $\mathbf{R}$).

Solution. Let $\varepsilon > 0$ and $a \in X$. Then we need to find $\delta > 0$ such that $d_X(x, a) < \delta \implies |f(x) - f(a)| < \varepsilon$. This is equivalent to

$$d_X(a, x_0) - \varepsilon < d_X(x, x_0) < d_X(a, x_0) + \varepsilon.$$

Since for any $x \in X$, $d_X(x, x_0) \leq d_X(x, a) + d_X(a, x_0)$ and $d_X(a, x_0) \leq d_X(a, x) + d_X(x, x_0)$ we can just take $\delta = \varepsilon$. ■

Exercise 1.

- a. Compute $d_{\sup}(x^2, x)$ in $\mathcal{C}[0, 1]$ and $d_1(x^2, x)$ in $\mathcal{L}_1[0, 1]$.
- b. Assume that $1 < a < b$ in $\mathbf{R}$. Compute $d_{\sup}(x^3, 2 - x^2)$ in $\mathcal{C}[a, b]$ and $d_1(x^3, 2 - x^2)$ in $\mathcal{L}_1[a, b]$.

Solution. a. On $[0, 1]$ we have $|x^2 - x| = x - x^2$. Moreover, $d(x - x^2)/dx = 1 - 2x$ is 0 if and only if $x = 1/2$. So $x = 1/2$ is a local extremum. Moreover $d^2(x - x^2)/dx^2 = -2 < 0$ when $x = 1/2$, so $x = 1/2$ is a local maximum. We conclude $d_{\sup}(x^2, x) = \sup_{x \in [0, 1]}(x - x^2) = 1/2 - 1/4 = 1/4$ in $\mathcal{C}[0, 1]$.

For d_1 a direct computation gives us $d_1(x^2, x) = \int_0^1 (x - x^2) dx = [x^2/2 - x^3/3]_0^1 = 1/2 - 1/3 - 0 = 1/6$ in $\mathcal{L}_1[0, 1]$.

- b. Since $1 < a < b$, we have $x^3 + x^2 > 2$ on $[a, b]$ and it attains its absolute maximum at $x = b$. So $d_{\sup}(x^3, 2 - x^2) = b^3 + b^2 - 2$ in $\mathcal{C}[a, b]$.

In $\mathcal{L}_1[a, b]$

$$d_1(x^3, 2 - x^2) = \int_a^b |x^3 + x^2 - 2| dx = \left[\frac{x^4}{4} + \frac{x^3}{3 - 2x} \right]_a^b,$$

which reduces to $\frac{b^4 - a^4}{4} + \frac{b^3 - a^3}{3} - 2(b - a)$. ■

Exercise 2.

Explain why neither of the following functions define a metric on the set of continuous functions $[0, 1] \rightarrow \mathbf{R}$:

- a. $d(f, g) = |f(1/2) - g(1/2)|$;
- b. $d(f, g) = \sup_{x \in [0, 1]} |f'(x) - g'(x)|$, where $f'(x) := df(x)/dx$.

Solution. a. Let $f(x) = x$ and $g(x) = 1 - x$. Then $d(f, g) = 0$ so axiom (M2) fails.

- b. If f or g fails to be differentiable for some $x \in [0, 1]$, then $d(f, g)$ is not defined! But even so, the examples $f(x) = 1$ and $g(x) = 0$ show that $d(f, g) = 0$ so (M2) fails again. ■

Exercise 3.

*Let $X := \{0, 1\}^\infty$ be the set of all infinite **binary sequences** $x = x_0x_1x_2 \dots$, where $x_n \in \{0, 1\}$, for $n \geq 0$; so $x = 100110111 \dots$ is a typical element. Show that the following are metrics on X .*

a.

$$d_{\min}(x, y) := \begin{cases} 0, & \text{if } x = y; \\ \frac{1}{2^n}, & \text{if } n = \min\{m \mid x_m \neq y_m\}. \end{cases}$$

b.

$$d^*(x, y) := \sum_{j=0}^{\infty} \frac{|x_j - y_j|}{2^j}.$$

Find two elements $x, y \in X$ such that $d_{\min}(x, y) \neq d^*(x, y)$.

Solution. a. (M1)–(M3) are satisfied automatically. For (M4), let $d_{\min}(x, y) = 1/2^p$ and $d_{\min}(y, z) = 1/2^q$. So $d_{\min}(x, z) = 1/2^r$, where $r = \min\{p, q\}$ if $p \neq q$ and $r > \min\{p, q\}$ if $p = q$. In both cases $1/2^r \leq 1/2^p + 1/2^q$, as required. If $x = y$ or $y = z$, the proof is trivial.

b. Notice that the right hand side is bounded above by 2 and therefore converges.

The axioms (M1)–(M4) then follow by elementary properties of convergent series. Taking $x = 0100 \dots$ and $y = 100 \dots$ and then $d_{\min}(x, y) = 1$ and $d^*(x, y) = 3/2$. ■

*Exercise 4.

Let X be the set of all closed intervals $[a, b]$ of $\mathbf{R}$. Show that

$$d([a, b], [c, d]) := \max\{|c - a|, |d - b|\}$$

is a metric on X .

Now consider the metric space (H, d_{∞}) , where

$$H := \{(x_1, x_2) \in \mathbf{R}^2 \mid x_1 \leq x_2\},$$

and show that it is isometric to (X, d) .

Solution. The axioms (M1) and (M2) are satisfied because $|c - a| = |d - b| = 0$ if and only if $c = a$ and $d = b$. Axiom (M3) is satisfied because $|x - y| = |y - x|$, for all $x, y \in \mathbf{R}$. For (M4), consider intervals $I = [a, b]$, $J = [c, d]$ and $K = [e, f]$ in X . Then

$$d(I, J) = \max\{|c - a|, |d - b|\} \leq \max\{|c - e| + |e - a|, |d - f| + |f - b|\},$$

since $|x - z| \leq |x - y| + |y - z|$, for all $x, y, z \in \mathbf{R}$. Also, $|c - e| + |e - a|$ and $|d - f| + |f - b|$ are both less than or equal to $\max\{|e - a|, |f - b|\} + \max\{|e - c|, |f - d|\}$ so

$$d(I, J) \leq \max\{|e - a|, |f - b|\} + \max\{|e - c|, |f - d|\} = d(I, K) + d(K, J).$$

We define a function $f: X \rightarrow H$ by $f([a, b]) = (a, b)$. It is easy to see that f is a bijection. Also, since

$$d_{\infty}((x_1, x_2), (y_1, y_2)) = \max\{|y_1 - x_1|, |y_2 - x_2|\},$$

f gives an isometry between (X, d) and (H, d_∞) . ■

Exercise 5.

Let $\mathcal{X}$ be the space of bounded sequences of real numbers. Show that the map $\|\cdot\|_\infty : \mathcal{X} \rightarrow \mathbf{R}$ defined by $\|a\|_\infty := \sup_{n \in \mathbf{N}} |a_n|$ is a norm on $\mathcal{X}$.

Solution. Before verifying the norm axioms, we need to check that the map $\|\cdot\|_\infty$ is well-defined. This is the case as a bounded implies that the supremum exists (and belongs to $\mathbf{R}$).

“(No)” It directly follows from the definition that $\|a\|_\infty \geq 0$ for all a .

“(N1) and (N2)” $\|a\|_\infty = 0$ if and only if $|a_n| = 0$ for all n , if and only if a is the zero sequence.

“(N3)” $\|\lambda a\|_\infty = \sup_{n \in \mathbf{N}} |\lambda a_n| = \sup_{n \in \mathbf{N}} |\lambda| |a_n| = |\lambda| \sup_{n \in \mathbf{N}} |a_n| = |\lambda| \|a\|_\infty$.

“(N4)” Let a and b be two sequences of real numbers. For every n we have $|a_n + b_n| \leq |a_n| + |b_n|$. This implies that for bounded sequences $\|a + b\|_\infty = \sup_{n \in \mathbf{N}} |a_n + b_n| \leq \sup_{n \in \mathbf{N}} |a_n| + \sup_{n \in \mathbf{N}} |b_n| = \|a\|_\infty + \|b\|_\infty$. ■

Exercise 6.

Let $\mathcal{Y}$ be the space of absolutely convergent sequences of real numbers. Show that the map $\|\cdot\|_1 : \mathcal{Y} \rightarrow \mathbf{R}$ defined by $\|a\|_1 := \sum_{n \in \mathbf{N}} |a_n|$ is a norm on $\mathcal{Y}$. (You may need to revise your properties of absolutely convergent series.)

Solution. By definition, $\mathcal{Y}$ is the space of sequences a such that $\|a\|_1$ is well-defined.

“(No)–(N2)” These are straightforward verifications, similar to Exercise 5

Before going further, let us recall the Rearrangement Theorem: The terms of an absolutely convergent series can be rearranged in any order, and the series will still converge to the same sum.

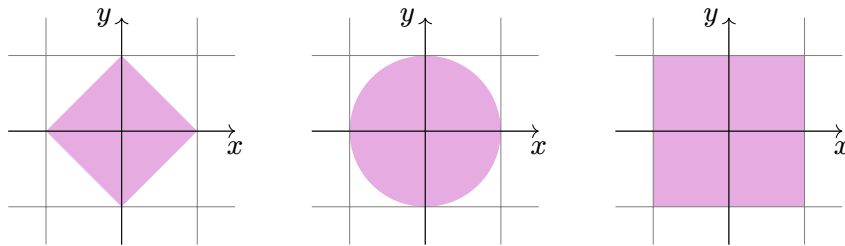
“(N3)” $\|\lambda a\|_1 = \sum_{n \in \mathbf{N}} |\lambda a_n| = \sum_{n \in \mathbf{N}} |\lambda| |a_n| = |\lambda| \sum_{n \in \mathbf{N}} |a_n| = |\lambda| \|a\|_1$.

“(N4)” Once again, for two sequences a and b of real numbers we have $|a_n + b_n| \leq |a_n| + |b_n|$ for every n . This implies that for absolutely convergent sequences $\|a + b\|_1 = \sum_{n \in \mathbf{N}} |a_n + b_n| \leq \sum_{n \in \mathbf{N}} (|a_n| + |b_n|) = \sum_{n \in \mathbf{N}} |a_n| + \sum_{n \in \mathbf{N}} |b_n| = \|a\|_1 + \|b\|_1$. ■

Exercise 7.

Sketch open balls of radius 1 around the origin in $\mathbf{R}^2$ with the different metrics d_1 , d_2 and d_∞ .

Solution.



For d_1 , the ball consists of all points (x, y) such that $|x + y| < 1$. The equation $|x + y| = 1$ gives us 4 straight segments (depending on the positivity of x and y) between the points $(0, 1)$, $(1, 0)$, $(0, -1)$ and $(-1, 0)$. The open ball is the interior of the square delimited by these segments.

For d_2 , the equation $1 = d((x, y), 0) = \sqrt{x^2 + y^2}$ defines a circle of radius 1 and the ball is the interior disk.

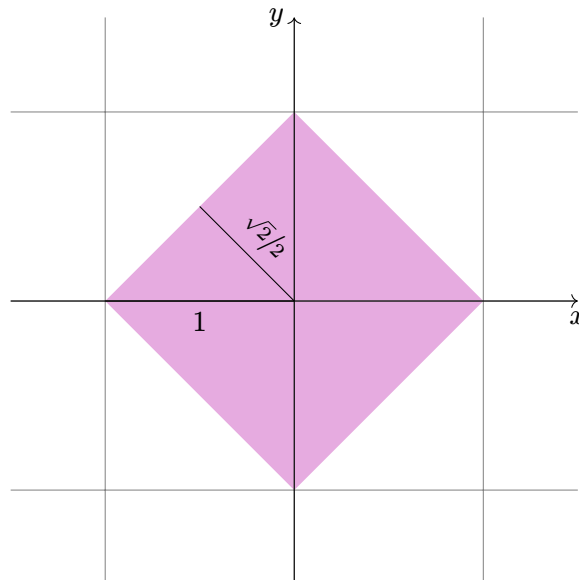
For d_∞ , the equation $1 = d((x, y), 0) = \max(x, y)$ defines a square with corners $(\pm 1, \pm 1)$.

Exercise 8.

Let $B_r(x)$ and $B'_r(x)$ denote open balls of radius r in $\mathbf{R}^2$ with respect to the Euclidean metric and the taxicab metric respectively. Find real numbers s and t such that

$$B_1(x) \subseteq B'_s(x) \subseteq B_{\sqrt{2}}(x) \subseteq B'_t(x) \subseteq B_2(x).$$

Solution. Each open ball $B'_r(x)$ has bounding line of greatest Euclidean distance r from x and least distance $r\sqrt{2}/2 = r/\sqrt{2}$.



Therefore,

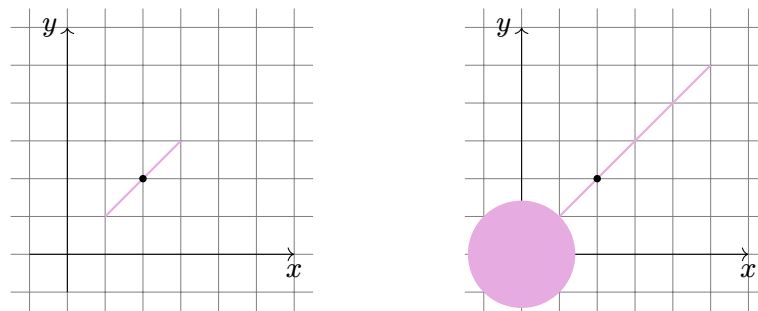
$$1 \leq \frac{s}{\sqrt{2}} < s \leq \sqrt{2} \leq \frac{t}{\sqrt{2}} < t \leq 2.$$

So the only possible solutions are $s = \sqrt{2}$ and $t = 2$. ■

Exercise 9.

Sketch open balls of radius $\sqrt{2}$ and $3\sqrt{2}$ around the point $(2, 2)$ in $\mathbf{R}^2$ with respect to the British Railway metric (see Tutorial 1 Exercise 5).

Solution.



If the radius r of the ball is smaller than the Euclidean distance from the center c to 0, then the balls consists of points p that are linearly dependent with c and of distance at most r to c . This is simply a line segment (on the line between 0 and c) of length $2r$ centred at c .

If $r \geq d_2(c, 0)$, then to the above line segment we can add $B_{r-d_2(c,0)}(0; d_2)$. Indeed, once we have reach 0 we can go back in any direction, but not further away than $r - d_2(c, 0)$.

Exercise 10.

Prove that if $x, y \in (X, d)$ are points in a metric space with $d(x, y) = 2r > 0$, then $B_r(x)$ and $B_r(y)$ are disjoint. Is it possible that in a metric space X containing more than one point that the only open sets are $\emptyset$ and X ?

Solution. Since $d(x, y) > 0$ we know that $x \neq y$. Suppose that $z \in B_r(x) \cap B_r(y)$. Then $d(x, y) \leq d(x, z) + d(z, y) < 2r$, which contradicts $d(x, y) = 2r$.

It is not possible since if $x, y \in X$ with $x \neq y$, then $d(x, y) = 2r > 0$, for some $r \in \mathbf{R}$. Then we know that $B_r(x) \cap B_r(y) = \emptyset$ and both of these open balls are non-empty and open in X . ■

To go further

The same idea shows that if X is a metric space containing at least n points, then it has at least n open sets different from $\emptyset$ and X .

Exercise 1.

Given any two subsets $U, V \subseteq (X, d)$ of a metric space show that

- $U \subseteq V \implies U^\circ \subseteq V^\circ$;
- $(U^\circ)^\circ = U^\circ$;
- U° is open in X ;
- U° is the largest subset of U that is open in X .

Solution. a. If $u \in U^\circ$, then $\exists \varepsilon > 0$ such that $B_\varepsilon(u) \subseteq U \subseteq V$; so $u \in V^\circ$;

b. By definition $(U^\circ)^\circ \subseteq U^\circ$. On the other hand, if $u \in U^\circ$, then $\exists \varepsilon > 0$ such that $B_\varepsilon(u) \subseteq U$, so $B_\varepsilon(u)^\circ \subseteq U^\circ$ by part a, but $B_\varepsilon(u)$ is open by [Lemma 1.5.9](#) and so $B_\varepsilon(u) = B_\varepsilon(u)^\circ$. Therefore, $u \in (U^\circ)^\circ$, so $U^\circ \subseteq (U^\circ)^\circ$.

c. This follows immediately from part b.

d. Suppose that $U^\circ \subseteq W \subseteq U$, where W is open in X . Then $W = W^\circ \subseteq U^\circ$ by part a. So $W \subseteq U^\circ$, and $U^\circ = W$. ■

Exercise 2.

Show that an open subset U of a metric space is a union of open balls.

Solution. Since U is open, for every $u \in U$, $\exists \varepsilon_u > 0$ such that $B_{\varepsilon_u}(u) \subseteq U$. Therefore, $\bigcup_{u \in U} B_{\varepsilon_u}(u) \subseteq U$. And since $u \in B_{\varepsilon_u}(u)$ we also have $U \subseteq \bigcup_{u \in U} B_{\varepsilon_u}(u)$. ■

Exercise 3.

Consider the set of binary sequences $\{0, 1\}^\infty$ with the metric d^* from [Tutorial 2 Exercise 3](#). What is the maximum possible value of $d^*(x, y)$, and for which pairs of sequences is this attained?

Describe the elements of the open ball $B_{3/4}(0)$ in $\{0, 1\}^\infty$, where 0 denotes the zero sequence $00 \dots$.

Solution. The maximum possible value of $d^*(x, y)$ is $\sum_{j=0}^{\infty} 1/2^j = 2$. It is attained by any pair of sequences which differ in every element; for example $d^*(0, 11 \dots) = 2$.

By definition, $B_{3/4}(0)$ contains all x such that $\sum_{j=0}^{\infty} 1/2^j < 3/4$. Thus $x_0 = 0$, otherwise $d^*(x, 0) \geq 1$. If $x_0 = 0$ and $x_1 = 1$, then $d^*(x, 0) = 1/2 + \sum_{j=2}^{\infty} 1/2^j$, so it is necessary that $x_2 = 0$ and $x_3 x_4 \dots \neq 11 \dots$. On the other hand, if $x_0 = x_1 = 0$, then $d^*(x, 0) \leq 1/2$.

So $x \in B_{3/4}(0)$ if and only if $x = 00z$ for any sequence z , or $x = 010w$ for any sequence $w \neq 11 \dots$. The additional elements in $\overline{B}_{3/4}(0)$ are $01011 \dots$ and $01100 \dots$. ■

Exercise 4.

On $\mathbf{R}^n$ check that

$$d_\infty(x, y) \leq d_2(x, y) \leq d_1(x, y) \leq n d_\infty(x, y).$$

where d_1 is the taxicab metric, d_2 is the Euclidean metric, and d_∞ is the max metric.

Solution. The first two inequalities follow from the definitions; you may find them easier to deduce by first considering the following observation:

$$(d_\infty(x, y))^2 \leq (d_2(x, y))^2 \leq (d_1(x, y))^2.$$

The final inequality again follows directly from the definitions. ■

Exercise 5.

Let (X, d_X) and (Y, d_Y) be two metric spaces and consider one of the metrics d_a, d_b, d_c on $X \times Y$.

a. Prove that the projection maps:

$$p_1: X \times Y \rightarrow X \quad \text{and} \quad p_2: X \times Y \rightarrow Y$$

given by $p_1(x, y) = x$ and $p_2(x, y) = y$ are continuous.

b. For some fixed $x_0 \in X$ and $y_0 \in Y$ prove that

$$i_{y_0}: X \rightarrow X \times Y \quad \text{and} \quad i_{x_0}: Y \rightarrow X \times Y$$

given by $i_{y_0}(x) = (x, y_0)$ and $i_{x_0}(y) = (x_0, y)$ are continuous.

Solution. Since d_a, d_b and d_c have the same open sets, continuity for d_a is equivalent to continuity for d_b , and also to continuity for d_c .

a. We will consider the metric d_a on $X \times Y$ and prove that p_1 is continuous. Let $(x, y) \in X \times Y$ and $\varepsilon > 0$. Then

$$d_X(p_1(x, y), p_1(x', y')) = d_X(x, x') \leq d_a((x, y), (x', y')) = d_X(x, x') + d_Y(y, y').$$

So we find that p_1 is continuous at $(x, y) \in X \times Y$ by taking $\delta = \varepsilon$. The proof for p_2 is symmetric.

b. We will prove that i_{y_0} is continuous taking the metric d_a again on $X \times Y$. Let $x \in X$ and $\varepsilon > 0$. Then

$$d_a(i_{y_0}(x), i_{y_0}(x')) = d_a((x, y_0), (x', y_0)) = d_X(x, x') + d_Y(y_0, y_0) = d_X(x, x').$$

So by taking $\delta = \varepsilon$ again we see that i_{y_0} is continuous at $x \in X$. The proof for p_2 is symmetric. ■

Exercise 6.

Given a metric space (X, d) , consider $(X \times X, d_a)$.

a. Prove that the diagonal map

$$\Delta: X \rightarrow X \times X$$

given by $\Delta(x) = (x, x)$ is continuous.

b. Show that the metric $d: (X \times X, d_a) \rightarrow (\mathbf{R}, d_{\mathbf{R}})$ is a continuous function.

Solution. a. Let $x \in X$ and $\varepsilon > 0$. Then

$$d_a(\Delta(x), \Delta(y)) = d_a((x, x), (y, y)) = 2d(x, y).$$

So we can take $\delta = \frac{\varepsilon}{2}$.

b. Let $(x, y) \in X \times X$ and $\varepsilon > 0$. Then we need to find $\delta > 0$ such that

$$d_a((x, y), (x', y')) < \delta \implies |d(x, y) - d(x', y')| < \varepsilon.$$

We have

$$\begin{aligned} |d(x, y) - d(x', y')| &\leq |(d(x, x') + d(x', y') + d(y', y)) - d(x', y')| \\ &= d(x, x') + d(y, y'). \end{aligned}$$

Since $d_a((x, y), (x', y')) = d(x, x') + d(y, y')$ we can take $\delta = \varepsilon$. ■

Exercise 7.

Suppose (X, d) is a metric space. Define $d'(x, y) = \min\{1, d(x, y)\}$. Prove that d' is a metric on X and then show that d and d' are topologically equivalent. Give an example to show that they may not be Lipschitz equivalent.

Solution. We have $d'(x, y) \leq d(x, y)$ and $0 < d'(x, y) \leq 1$, for all $x \neq y \in X$. So for the triangle inequality notice that

$$\begin{aligned} d'(x, z) &= \min\{1, d(x, z)\} \leq \min\{1, d(x, y) + d(y, z)\} \\ &\leq \min\{1, d(x, y)\} + \min\{1, d(y, z)\} = d'(x, y) + d'(y, z). \end{aligned}$$

In terms of open balls, notice that

$$B_r(x; d') = \begin{cases} B_r(x; d), & \text{if } r \leq 1; \\ X, & \text{if } r > 1, \end{cases}$$

so $B_r(x; d) \subseteq B_r(x; d')$, for all $x \in X$ and $r > 0$. Therefore, it automatically follows that if $U \subseteq X$ is d' -open, then it is d -open. Suppose now that $U \subseteq X$ is d -open and

let $u \in U$. Then $\exists \varepsilon > 0$ such that $B_\varepsilon(u; d) \subseteq U$. Let $\varepsilon' := \min\{1, \varepsilon\} \leq 1$. Then $B_{\varepsilon'}(u; d') \subseteq B_\varepsilon(u; d) \subseteq U$ so U is d' -open.

Obviously, for any metric space (X, d) , we find that (X, d') is bounded. But an unbounded metric space cannot be Lipschitz equivalent to a bounded one by [Lemma 1.9.4](#). For a concrete counterexample, take $X = \mathbf{R}$ and $d = d_{\mathbf{R}}$ its standard metric. If d and d' were Lipschitz equivalent, then there would exist $h, k \in (0, \infty)$ such that

$$hd'(x, y) \leq d(x, y) = |x - y| \leq d'(x, y) \leq 1, \quad \forall x, y \in \mathbf{R}.$$

Taking x and y to be further than distance 1 away from each other, for example, $x = 0, y = 2$, we find a contradiction. ■

Exercise 8.

Prove that isometries and Lipschitz equivalences between metric spaces are continuous.

Solution. Isometries are continuous by taking $\delta = \varepsilon$ in the definition of continuity. A Lipschitz equivalence $f: (X, d_X) \rightarrow (Y, d_Y)$ (using the same notation as in its definition) is continuous at $x \in X$ by taking $\delta = h\varepsilon$ since:

$$d_Y(f(x), f(y)) \leq \frac{1}{h}d_X(x, y) < \frac{\delta}{h} = \varepsilon. \quad \blacksquare$$

Exercise 9.

Check that being an isometry defines an equivalence relation on all metric spaces. Do the same for Lipschitz equivalences and homeomorphisms.

Solution. “ $X \sim X$ ” The identity map $\text{Id}: X \rightarrow X$ is bijective and an isometry as well as being its own inverse. It is thus a Lipschitz equivalence and a continuous map.

“ $X \sim Y \Rightarrow Y \sim X$ ” If $f: X \rightarrow Y$ is an isometry, respectively a homeomorphism, then $f^{-1}: Y \rightarrow X$ is automatically an isometry, respectively a homeomorphism. If f is a Lipschitz equivalence, with parameters h and k , then

$$hd_Y(f(x), f(y)) \leq d_X(x, y) \leq kd_Y(f(x), f(y)).$$

By letting $w = f(x)$ and $z = f(y)$ we have

$$hd_Y(w, z) \leq d_X(f^{-1}(w), f^{-1}(z)) \leq kd_Y(w, z),$$

which is equivalent to

$$\frac{1}{k}d_X(f^{-1}(w), f^{-1}(z)) \leq d_Y(w, z) \leq \frac{1}{h}d_Y(w, z).$$

So f^{-1} is also a Lipschitz equivalence. “ $X \sim Y, Y \sim Z \Rightarrow X \sim Z$ ” This follows directly from the composition of continuous functions. ■

Exercise 10.

In the proof that two Lipschitz equivalent metrics d and d' on X are topologically equivalent ([Proposition 1.7.4](#)) prove the statements regarding open balls.

Solution. Let $y \in B_{h\varepsilon}(x; d)$. Then $d(x, y) < h\varepsilon$, which implies that $d'(x, y) \leq \frac{1}{h}d(x, y) < \varepsilon$. Similarly, if $y \in B_{\varepsilon/k}(x; d')$, then $d(x, y) \leq kd'(x, y) < \varepsilon$. ■

Exercise 1.

Prove that the *open annulus* $\{x \in \mathbf{R}^2 \mid 4 < x_1^2 + x_2^2 < 9\}$ and the *punctured plane* $\mathbf{R}^2 \setminus \{0\}$ are homeomorphic subspaces of $\mathbf{R}^2$.

Solution. The annulus $A := \{x \in \mathbf{R}^2 \mid 4 < x_1^2 + x_2^2 < 9\}$ lies between the circles of radius 2 and 3, so every $a \in A$ may be written as $(r \cos \theta, r \sin \theta)$ for some $r \in (2, 3) \subseteq \mathbf{R}$. By [Example 1.7.3](#), there exists a homeomorphism $t: (2, 3) \rightarrow (0, +\infty)$ (for example, one can take $t(x) = \tan(x\pi/2)$). Define

$$h: A \rightarrow \mathbf{R}^2 \setminus \{0\} \quad \text{by} \quad h(a) = (t(r) \cos \theta, t(r) \sin \theta).$$

This is continuous by the composition of continuous functions and it has a continuous inverse $h^{-1}(s \cos \theta, s \sin \theta) = (t^{-1}(s) \cos \theta, t^{-1}(s) \sin \theta)$. For our concrete example, one have $t^{-1}(s) = \frac{2}{\pi} \tan^{-1} s$. ■

Exercise 2.

Let $\ell^1 = (\mathcal{Y}, d_1)$ be the sequence space of absolutely convergent sequences (see [Definition 1.4.3](#)) with the d_1 metric.

Find a distance preserving injection of the space $(\{0, 1\}^\infty, d^*)$ (see [Tutorial 3 Exercise 4](#)) into ℓ^1 .

Find an example of a point $a \in \mathcal{Y}$ such that $a \in \overline{B}_{1/10}(0)$ but $a \notin B_{1/10}(0)$.

Solution. Consider the function $i: \{0, 1\}^\infty \rightarrow \mathcal{Y}$ defined by

$$i(x_0 x_1 \dots x_k \dots) = (x_0, x_1/2, \dots, x_k/2^k, \dots);$$

the right hand side is well defined, because

$$\sum_{k \in \mathbf{N}} \left| \frac{x_k}{2^k} \right| \leq \sum_{k \in \mathbf{N}} \frac{1}{2^k} = 2,$$

as required. Obviously i is injective; it also preserves distance, because

$$d^*(x, y) = \sum_{k \in \mathbf{N}} \frac{|x_k - y_k|}{2^k} = \sum_{k \in \mathbf{N}} \left| \frac{x_k}{2^k} - \frac{y_k}{2^k} \right| = d_1(i(x), i(y)).$$

A point $a \in \mathcal{Y}$ satisfies the condition if and only if $\sum_{k=0}^{\infty} |a_k| = 1/10$. Any example, such as

$$\left(\frac{1}{10}, 0, 0, \dots \right) \quad \text{or} \quad \left(\frac{1}{20}, \frac{1}{40}, \dots, \frac{1}{10 \cdot 2^{j+1}}, \dots \right)$$

will do. ■

Exercise 3.

Given any two subsets $U, V \subseteq X$ prove the following:

- $U \subseteq V$ implies $\overline{U} \subseteq \overline{V}$;
- $\overline{\overline{V}} = \overline{V}$;
- $\overline{V}$ is closed in X ;
- $\overline{V}$ is the smallest set containing V that is closed in X .

Solution. a. For every $\varepsilon > 0$, if $B_\varepsilon(x) \cap U$ is non-empty, so is $B_\varepsilon(x) \cap V$.

b. By definition, $\overline{V} \subseteq \overline{\overline{V}}$. Conversely, if $x \in \overline{\overline{V}}$, then $B_{\varepsilon/2}(x) \cap \overline{V}$ is non-empty for every $\varepsilon > 0$. So there exists w for which $d(x, w) < \varepsilon/2$ and $w \in \overline{V}$, which implies that $B_{\varepsilon/2}(w) \cap V$ is non-empty; therefore $B_\varepsilon(x) \cap V$ is also non-empty, by the triangle inequality. Hence $x \in \overline{V}$ and $\overline{\overline{V}} \subseteq \overline{V}$ as required.

c. This follows immediately from b.

d. Suppose that $V \subseteq W \subseteq \overline{V}$, where W is closed. Then $\overline{V} \subseteq \overline{W} = W$ by a. So $\overline{V} \subseteq W$, and $W = \overline{V}$ as required. ■

Exercise 4.

Determine whether the following subsets are open, closed, or neither in $\mathbf{R}$ (with the standard metric):

- $[0, 1)$;
- $\mathbf{Z}$;
- $\{x \in \mathbf{R} \mid \sin x > 0\}$;
- $\bigcup_{n=2}^{\infty} [1/n, 1)$.

Solution. a. $[0, 1)$ is not open since 0 does not lie in the interior. It is not closed because 1 is a closure point that does not belong to the set.

b. $\mathbf{Z}$ is closed in $\mathbf{R}$ because its complement is open as a union of open intervals.

c. $\{x \in \mathbf{R} \mid \sin x > 0\} = \bigcup_{n \in \mathbf{Z}} (2n\pi, (2n+1)\pi)$ is a union of open intervals in $\mathbf{R}$, and is therefore open.

d. $\bigcup_{n=2}^{\infty} [1/n, 1) = (0, 1)$ is open in $\mathbf{R}$. ■

Exercise 5.

For a subset $W \subseteq (X, d)$ of a metric space prove that

- a. $W^\circ = X \setminus \overline{(X \setminus W)}$;
- b. $\overline{W} = X \setminus (X \setminus W)^\circ$.

Solution. a. $w \in W^\circ \iff \exists \varepsilon > 0 : B_\varepsilon(w) \subseteq W \iff \exists \varepsilon > 0 : B_\varepsilon(w) \cap (X \setminus W) = \emptyset$
 $\iff w \notin \overline{X \setminus W}$.

b. $x \in \overline{W} \iff \forall \varepsilon > 0 : B_\varepsilon(x) \cap W \neq \emptyset \iff \forall \varepsilon > 0 : B_\varepsilon(x) \not\subseteq X \setminus W$
 $\iff x \notin (X \setminus W)^\circ$. ■

Exercise 6.

In a metric space (X, d) prove that every open ball is contained in a closed ball and that every closed ball is contained in an open ball. Is it always true that $\overline{B_r(x)}$ is $\overline{B_r(x)}$? Give a proof or a counterexample.

Solution. For any $x \in X$ and $r > 0$,

$$B_r(x) \subseteq \overline{B_r(x)} \subseteq B_{2r}(x).$$

It is not true that the closure of $B_r(x)$ is always $\overline{B_r(x)}$. In a discrete metric space (X, d) we the ball $B_1(x) = \{x\}$ is both open and close, so $\overline{B_1(x)} = \{x\}$ as well. But $\overline{B_1(x)} = X$, which is not the same as $\{x\}$ if X contains more than one point. ■

Exercise 7.

Consider a subset $W \subseteq (X, d)$ of a metric space. Recall that a point $x \in X$ is an accumulation point of W if, for every $\varepsilon > 0$, $B_\varepsilon(x) \cap W \setminus \{x\} \neq \emptyset$. Let $\text{Acc}(W)$ denote the set of all accumulation points of W . Prove that $\overline{W} = W \cup \text{Acc}(W)$. Recall that $w \in W$ is an isolated point of W if there exists $\varepsilon > 0$ such that $B_\varepsilon(w) \cap W = \{w\}$. Let $\text{Iso}(W)$ denote the set of all isolated points of W . Prove that $\text{Iso}(W) = W \setminus \text{Acc}(W)$.

Solution. Let $x \in \overline{W}$. Then if $x \notin W$ we have have that $B_\varepsilon(x) \cap W \setminus \{x\} \neq \emptyset$, for every $\varepsilon > 0$. So $x \in \text{Acc}(W)$. It is easy to see that if $x \in \text{Acc}(W)$, then $x \in \overline{W}$.

Let $w \in \text{Iso}(W)$. Then $w \in W$ by definition and $B_\varepsilon(w) \cap W = \{w\}$, for some $\varepsilon > 0$, which implies that $B_\varepsilon(w) \cap W \setminus \{w\} = \emptyset$ so $w \notin \text{Acc}(W)$. Suppose that $w \in W \setminus \text{Acc}(W)$. Then there is an $\varepsilon > 0$ such that $B_\varepsilon(w) \cap W \setminus \{w\} = \emptyset$ which implies that $B_\varepsilon(w) \cap W = \{w\}$ so $w \in \text{Iso}(W)$. ■

Exercise 1.

Prove that if $Y \subseteq Z$, where Z is a bounded subset of (X, d) , then Y is bounded and $\text{diam } Y \leq \text{diam } Z$.

Solution. Since Z is bounded we know there exists $x \in X$ and $r > 0$ such that $Z \subseteq \overline{B}_r(x)$ and since $Y \subseteq Z$ we have that Y is bounded. Also, for each $y_1, y_2 \in Y \subseteq Z$, $d(y_1, y_2) \leq \text{diam } Z$ so $\text{diam } Z$ is an upper bound for $\{d(y_1, y_2) \mid y_1, y_2 \in Y\}$ which implies that $\text{diam } Y := \sup\{d(y_1, y_2) \mid y_1, y_2 \in Y\} \leq \text{diam } Z$. ■

Exercise 2.

Prove that if Y, Z are bounded subsets of (X, d) with $Y \cap Z \neq \emptyset$, then

$$\text{diam } Y \cup Z \leq \text{diam } Y + \text{diam } Z.$$

Solution. If $u_1, u_2 \in Y \cup Z$ both belong to Y , then

$$d(u_1, u_2) \leq \text{diam } Y \leq \text{diam } Y + \text{diam } Z.$$

If u_1, u_2 both belong to Z , a interchanging the roles of Y and Z gives the desired result. If $u_1 \in Y$ and $u_2 \in Z$ (or the the other way around) then take some $\alpha \in Y \cap Z$ and

$$d(u_1, u_2) \leq d(u_1, \alpha) + d(\alpha, u_2) \leq \text{diam } Y + \text{diam } Z.$$

Therefore, $d(u_1, u_2) \leq \text{diam } Y + \text{diam } Z$, for all $u_1, u_2 \in Y \cup Z$, which gives the result. ■

Exercise 3.

For a subset $W \subseteq (X, d)$ of a metric space, prove that $\partial W = \overline{W} \cap \overline{(X \setminus W)}$.

Solution. $x \in \partial W \iff x \in \overline{W} \setminus W^\circ \iff (x \in \overline{W} \& \forall \varepsilon > 0 : B_\varepsilon(x) \cap (X \setminus W) \neq \emptyset) \iff (x \in \overline{W} \& x \in \overline{(X \setminus W)})$. ■

Exercise 4.

Determine ∂S , S° , and $\overline{S}$ for each of the subspaces $S := [0, 1) \cup (2, 3]$, $S := \mathbf{Z}$, and $S := \mathbf{R} \setminus \mathbf{Q}$ of the Euclidean line.

Solution.

$$\begin{aligned} \partial[0, 1) \cup (2, 3] &= \{0, 1, 2, 3\}, & ([0, 1) \cup (2, 3])^\circ &= (0, 1) \cup (2, 3), \\ \overline{[0, 1) \cup (2, 3]} &= [0, 1] \cup [2, 3]. \end{aligned}$$

$$\partial \mathbf{Z} = \mathbf{Z}, \quad (\mathbf{Z})^\circ = \emptyset, \quad \overline{\mathbf{Z}} = \mathbf{Z}.$$

$$\partial(\mathbf{R} \setminus \mathbf{Q}) = \mathbf{R}, \quad (\mathbf{R} \setminus \mathbf{Q})^\circ = \emptyset, \quad \overline{(\mathbf{R} \setminus \mathbf{Q})} = \mathbf{R}. \quad \blacksquare$$

Exercise 5.

Is $\mathbf{Q}^n$ dense in $\mathbf{R}^n$? Let $\mathbf{Z}[\frac{1}{2}] := \{\frac{m}{2^n} \mid m, n \in \mathbf{Z}\}$. Is $\mathbf{Z}[\frac{1}{2}]$ dense in $\mathbf{R}$? Is $\mathbf{Z}[\frac{1}{2}]^n$ dense in $\mathbf{R}^n$.

Solution. To prove that $\mathbf{Q}^n$ is dense in $\mathbf{R}^n$ we must prove that $\overline{\mathbf{Q}^n} = \mathbf{R}^n$. It is sufficient to show that every $(x_1, \dots, x_n) \in \mathbf{R}^n$ is the limit of a sequence in $\mathbf{Q}^n$. But we know that $\mathbf{Q}$ is dense in $\mathbf{R}$, so there exists a sequence $(q_{i,k})_{k \geq 1}$ converging to x_i for every $1 \leq i \leq n$. Then $(q_k = (q_{1,k}, \dots, q_{n,k}))_{k \geq 1}$ converges to x , because

$$d(x, q_k) = \left(\sum_{i=1}^n (x_i - q_{i,k})^2 \right)^{1/2} \rightarrow 0, \quad \text{as } k \rightarrow \infty.$$

Given any $z \in \mathbf{R}$, let z_n denote the truncation of its binary expansion after n digits, for every $n \geq 1$. So $z_n \in \mathbf{Z}[\frac{1}{2}]$ by construction, and $z_n \rightarrow z$ as $n \rightarrow \infty$. So $\mathbf{Z}[\frac{1}{2}]$ is dense in $\mathbf{R}$. Using a similar argument as for $\mathbf{Q}^n$ shows that $\mathbf{Z}[\frac{1}{2}]^n$ is dense in $\mathbf{R}^n$. ■

Exercise 6.

Show that a subset $U \subseteq X$ of a metric space is open if and only if for every sequence (x_n) of X that converges to a point in U , there exists $K \in \mathbf{N}$ such that $x_k \in U$, for all $k \geq K$.

Solution. “ $\Rightarrow$ ” Suppose that U is open in X and that (x_n) is a sequence in X that converges to $u \in U$. Since U is open, there exists $\varepsilon > 0$ such that $B_\varepsilon(u) \subseteq U$, and for this $\varepsilon > 0$, $\exists K \in \mathbf{N}$ such that $k \geq K \implies d(x_k, u) < \varepsilon$, i.e. $x_k \in B_\varepsilon(u) \subseteq U$, for all $k \geq K$.

“ $\Leftarrow$ ” Let $u \in U$ and suppose $u \notin U^\circ$, that is, for every $\varepsilon > 0$, $B_\varepsilon(u) \cap (X \setminus U) \neq \emptyset$. Choose $x_i \in B_{1/i}(u) \cap (X \setminus U)$. Then (x_n) is a sequence in X such that $\lim_n x_n = u$ but $x_k \notin U$, for all $k \in \mathbf{N}$. ■

Exercise 7.

Let $\mathcal{P}[0, 1] \subseteq \mathcal{C}[0, 1]$ be the subspace consisting of all polynomials functions. By constructing the sequence

$$f_n(x) := 1 + \frac{x}{2} + \frac{x^2}{4} + \dots + \frac{x^n}{2^n} \in \mathcal{P}[0, 1],$$

show that $\mathcal{P}[0, 1]$ is not closed in $\mathcal{C}[0, 1]$.

Solution. Let $f(x) = (1 - x/2)^{-1}$ in $\mathcal{C}[0, 1]$; then the Binomial Theorem implies that $f(x) - f_n(x) = \sum_{k \geq n+1} x^k / 2^k \leq 1/2^n$ for $x \in [0, 1]$. So

$$d_{\text{sup}}(f_n, f) = \sup_{x \in [0, 1]} |f_n(x) - f(x)| = 1/2^n \rightarrow 0, \quad \text{as } n \rightarrow \infty,$$

and $f_n \rightarrow f$ in $\mathcal{C}[0, 1]$. But f is not a polynomial, so $\mathcal{P}[0, 1]$ is not closed. ■

Exercise 8.

Given the sequence space $(\{0,1\}^\infty, d^*)$ (see *Tutorial 3 Exercise 4*), consider $f_1: \{0,1\}^\infty \rightarrow \{0,1\}^\infty$ defined by inserting 1 at the beginning of every sequence. Prove that f_1 is continuous and decreases the distance in the sense that $d^*(f_1(x), f_1(y)) \leq d^*(x, y)$.

Solution. By definition $f_1(x) = 1x_0x_1 \dots$ and $f_1(y) = 1y_0y_1 \dots$. So

$$d^*(f_1(x), f_1(y)) = \sum_{j=1}^{\infty} \frac{|x_{j-1} - y_{j-1}|}{2^j} = \frac{1}{2} \sum_{j=1}^{\infty} \frac{|x_{j-1} - y_{j-1}|}{2^{j-1}} = \frac{1}{2} d^*(x, y).$$

In particular, $d^*(f_1(x), f_1(y)) \leq d^*(x, y)$, with equality only if $x = y$.

It also follows that $B_{2\varepsilon}(x) = B_\varepsilon(f_1(x))$ for any $\varepsilon > 0$, and therefore that f_1 is continuous by choosing $\delta \leq 2\varepsilon$. ■

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

Which of the following subsets of $\mathbf{R}^2$ are connected? Which are path-connected?

- $B_1((1,0)) \cup B_1((-1,0));$
- $\overline{B_1}((1,0)) \cup \overline{B_1}((-1,0));$
- $\overline{B_1}((1,0)) \cup B_1((-1,0));$
- The “rational comb” $C := \{(q, y) \in \mathbf{R}^2 \mid q \in \mathbf{Q}, y \in [0, 1]\} \cup (\mathbf{R} \times \{1\});$
- The set of all points in $\mathbf{R}^2$ with at least one coordinate in $\mathbf{Q}$.

- Solution.*
- This is a disjoint union of two open sets. It is thus not connected and therefore not path-connected.
 - This is a union of two path-connected sets (the two balls) with non-empty intersection (it contains $(0,0)$). It is therefore path-connected, and hence connected.
 - The two balls are path-connected. For x in $B_1((-1,0))$ the line segment between 0 and x stays inside $\overline{B_1}((1,0)) \cup B_1((-1,0))$ and is thus a path from 0 to x . We conclude that $\overline{B_1}((1,0)) \cup B_1((-1,0))$ is path-connected, and hence connected.
 - A point of the form (q, y) ($q \in \mathbf{Q}, y \in [0, 1]$) is connected by a (vertical) path to $(q, 1)$, which is connected by a (horizontal path) to $(0, 1)$. All points of the form $(r, 1)$ are connected by a path to $(0, 1)$. We conclude that C is path-connected, and hence connected.
 - Any point (p, r) with $p \in \mathbf{Q}$ is connected by a vertical path to $(p, 0)$, which is connected to $(0, 0)$ by a horizontal path. Similarly, any point (r, q) with $q \in \mathbf{Q}$ is path-connected. We conclude that our set is path-connected, and hence connected. ■

Exercise 2.

Suppose that $A \mid B$ is a partition of a metric space X and that $f: X \rightarrow Y$ is a function to another metric space Y . Prove that if the restrictions $f|_A$ and $f|_B$ are both continuous, then f is continuous.

Solution. Let U be an open subset of Y . Set $U_A := f^{-1}(U) \cap A$ and $U_B := f^{-1}(U) \cap B$. Since both $f|_A$ and $f|_B$ are continuous we find that $U_A = (f|_A)^{-1}(U)$ is open in A and $U_B = (f|_B)^{-1}(U)$ is open in B . Since $A \mid B$ is a partition of X both U_A and U_B are also open in X . Therefore, $f^{-1}(U) = U_A \cup U_B$ is open in X . ■

Exercise 3.

Suppose that for each $i \in \{1, \dots, n\}$ that A_i is a connected subspace of a metric space X , such that $A_i \cap A_{i+1} \neq \emptyset$ for each $i \in \{1, \dots, n-1\}$. Prove that $\bigcup_{i=1}^n A_i$ is connected. Does this result extend to an infinite sequence (A_i) of connected subsets?

Solution. Suppose $f: \bigcup_{i=1}^n A_i \rightarrow \{0, 1\}$ is continuous. The restriction $f|_{A_i}: A_i \rightarrow \{0, 1\}$ is continuous and therefore constant since A_i is connected. Let us write c_i for the value $f|_{A_i}(x)$ (for any $x \in A_i$). Since the intersection $A_i \cap A_{i+1}$ is not empty, then $c_i = c_{i+1}$ for all i . We conclude that all the c_i are equal and thus that f is constant.

The infinite case is also true, with the same proof. ■

Exercise 4.

Show that a metric space X is connected if and only if for every $W \subseteq X$ such that $W \neq \emptyset, X$, the boundary ∂W is non-empty.

Solution. A metric space X is connected if and only if the only subsets which are both open and closed are $\emptyset$ and X . But W is both open and closed if and only if $\overline{W} = W^\circ$, if and only if $\partial W = \emptyset$. ■

Exercise 5.

If X is a connected metric space containing more than one point, and if $\{x\}$ is closed for every $x \in X$, show that X is infinite.

Solution. If X is finite and all points are closed, then $X \setminus \{x\}$ is a finite union of closed sets and is therefore closed. Therefore, $\{x\} = X \setminus (X \setminus \{x\})$ is both open and closed and X cannot be connected (unless $X = \{x\}$). ■

Exercise 6.

Prove that any continuous function $f: [a, b] \rightarrow [a, b]$ has a fixed point, that is, a point x such that $f(x) = x$.

Hint: Consider the function $g: [a, b] \rightarrow \mathbf{R}$ given by $g(x) = f(x) - x$.

Solution. The function g in the hint is a continuous function from a closed interval to $\mathbf{R}$ and, therefore, its image is a closed interval. Since $a \leq f(x) \leq b$, for all $x \in [a, b]$, we have

$$0 \leq g(a) = f(a) - a \leq b - a \quad \text{and} \quad a - b \leq g(b) = f(b) - b \leq 0.$$

So if $f(a) \neq a$, then $g(a) > 0$, and if $f(b) \neq b$, then $g(b) < 0$. So by the Intermediate Value Theorem, there exists $x \in (a, b)$ such that $g(x) = 0$, i.e. $f(x) = x$. ■

Exercise 7.

- Prove that if X is a connected metric space and there exists a non-constant continuous function $f: X \rightarrow \mathbf{R}$, then X is uncountable.
- Prove that a connected metric space containing more than one point contains uncountably many points.

Solution. a. If f takes two real values $a < b$, then by the Intermediate Value Theorem, it takes on all values in $[a, b]$, which is uncountable, and hence so is X .

b. Apply part a to $f: X \rightarrow \mathbf{R}$ given by $f(x) = d(x, a)$ for a fixed $a \in X$. ■

Exercise 8.

Show that there is no continuous injective function $f: \mathbf{R}^2 \rightarrow \mathbf{R}$.

Hint: Consider a triangle in $\mathbf{R}^2$.

Solution. Consider a triangle ABC in $\mathbf{R}^2$. Then BC and $BA \cup AC$ are both connected. So both $f(BC)$ and $f(BA \cup AC)$ contain $[f(B), f(C)]$ by the Intermediate Value Theorem. ■

***Exercise 9.**

Suppose that $f: [a, b] \rightarrow \mathbf{R}$ is a function such that $|f^{-1}(y)| = 2$, for every y in the image of f . Prove that f cannot be continuous.

Solution. For every $x \in [a, b]$ there exists a unique $x' \in [a, b]$, with $x \neq x'$, such that $f(x) = f(x')$. We define a function that switches these points:

$$\begin{aligned} g: [a, b] &\longrightarrow [a, b] \\ x &\longmapsto g(x) = f^{-1}(f(x)) \setminus \{x\}. \end{aligned}$$

If f is continuous, $x \in [a, b]$, and (x_n) is a sequence in $[a, b]$ such that $\lim x_n = x$, then

$$\lim g(x_n) = \lim (f^{-1}(f(x_n)) \setminus \{x_n\}) = g(x)$$

since $\lim f(x_n) = f(x)$. So g is continuous but has no fixed point, contradicting Exercise 6. ■

Exercise 10.

- Suppose that $f: X \rightarrow Y$ is a continuous function between metric spaces and that X is path-connected. Show that $f(X)$ is path-connected.
- Suppose that X and Y are homeomorphic metric spaces. Show that X is path-connected if and only if Y is path-connected.

c. Show that the graph G_f of a continuous function with path-connected domain is path-connected.

- Solution.* a. Let $y_1, y_2 \in f(X)$. Then there exists $x_1, x_2 \in X$ such that $y_1 = f(x_1)$ and $y_2 = f(x_2)$. Since X is path-connected, there exists a path $g: I \rightarrow X$ from x_1 to x_2 . The composition $f \circ g: I \rightarrow Y$ is continuous, as a composition of continuous functions, and therefore gives a path in Y from y_1 to y_2 .
- b. Suppose that $f: X \rightarrow Y$ is a homeomorphism. By the first part, if X is path-connected then $f(X) = Y$ is path-connected. Similarly, if Y is path-connected then $X = f^{-1}(Y)$ is path-connected.
- c. Recall that the graph G_f of a continuous function is homeomorphic to its domain. Therefore, if the domain is path-connected, then G_f is path-connected by the first part. ■

***Exercise 11.**

Suppose $f, g: I \rightarrow X$ are paths in a metric space X such that $f(1) = g(0)$. Show that

$$h(x) := \begin{cases} f(2x), & x \in [0, 1/2]; \\ g(2x - 1), & x \in [1/2, 1], \end{cases}$$

is a path in X from $f(0)$ to $g(1)$.

Solution. The map h is well defined, $h(0) = f(0)$ and $h(1) = g(1)$. We hence need to show that it is continuous.

First, observe that if $f: [a, b] \rightarrow X$ is continuous, so are $f_r: [a + r, b + r] \rightarrow X$ and $f_\lambda: [\lambda a, \lambda b] \rightarrow X$ for any $r \in \mathbf{R}$ and $0 \neq \lambda \in \mathbf{R}$, where $f_r(x) = f(x - r)$ and $f_\lambda(x) = f(\frac{\lambda}{x})$.

Using the above observation, we want to show that if $f: [0, 1] \rightarrow \mathbf{R}$ and $g: [1, 2] \rightarrow \mathbf{R}$ are continuous, then so is the map $h: [0, 2] \rightarrow \mathbf{R}$ defined by $h(x) = f(x)$ if $x \leq 1$ and $h(x) = g(x)$ if $x \geq 1$. Let U be an open subset of X . Since $h[0, 2] = f[0, 1] \cup g[1, 2]$ we have $h^{-1}(U) = f^{-1}(U) \cup g^{-1}(U)$. The subset $f^{-1}(U)$ is open in $[0, 1]$ by continuity of f . It is thus a union of balls, that is of the form $\bigcup_{i \in I} (B_i \cap [0, 1])$, where the B_i are open intervals of $\mathbf{R}$. This implies that $f^{-1}(U)$ is open in $[0, 2]$ if and only if it does not contain 1. A similar statement is true for $g^{-1}(U) = \bigcup_{j \in J} (C_j \cap [1, 2])$ for some open intervals C_j . If none of $f^{-1}(U)$ and $g^{-1}(U)$ contains 1, they are both open in $[0, 2]$ and so is their union. If one of $f^{-1}(U)$ or $g^{-1}(U)$ contains 1 they both do and therefore therefore, for every B_i containing 1 there exists C_j (also containing 1) such that $(B_i \cup C_j) \cap [0, 2]$ is open; and vice-versa for C_j and B_i . This implies that $f^{-1}(U) \cup g^{-1}(U)$ is open in $[0, 2]$. ■

***Exercise 12.**

- a. Let X be a metric space and show that: “ $x \sim y$ if there is a path in X from x to y ” is an equivalence relation.
- b. Let $\pi(X)$ denote the set of equivalence classes (also known as connected components). For any continuous function $f: X \rightarrow Y$ we define a function

$$\pi(f): \pi(X) \rightarrow \pi(Y)$$

by: if $C(x) \in \pi(X)$ is the connected component containing $x \in X$, then

$$\pi(f)(C(x)) := C(f(x)).$$

Show that $\pi(f)$ is a well-defined function.

- c. Given continuous functions $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ show that the following diagram commutes:

$$\begin{array}{ccc} \pi(X) & \xrightarrow{\pi(f)} & \pi(Y) \\ & \searrow \pi(g \circ f) & \swarrow \pi(g) \\ & \pi(Z) & \end{array}$$

- d. Show that if $\text{Id}: X \rightarrow X$ is the identity map of X , then $\pi(\text{Id}): \pi(X) \rightarrow \pi(X)$ is the identity map of $\pi(X)$.
- e. Show that if $f: X \rightarrow Y$ is a homeomorphism, then $\pi(f): \pi(X) \rightarrow \pi(Y)$ is a bijection.

Solution. a. The constant path $f: I \rightarrow X$, $f(t) = x$ for all $t \in I$, shows that $x \sim x$. If $f: I \rightarrow X$ is a path from x to y , then $g: I \rightarrow X$, $g(t) = f(1 - t)$, gives a path from y to x ; so $x \sim y \iff y \sim x$. Finally, *Exercise 11 shows that $x \sim y$ & $y \sim z \implies x \sim z$.

- b. Suppose that $y \in C(x)$, i.e. $x \sim y$. We need to show that $C(f(x)) = C(f(y))$. Since $x \sim y$, we know that $C := C(x) = C(y)$, where C is a connected component of X . So $f(x), f(y) \in f(C)$ and $f(C)$ is the continuous image of a path-connected space and is therefore path-connected. Hence $C(f(x)) = C(f(y))$.

- c. For any $x \in X$,

$$\begin{aligned} (\pi(g) \circ \pi(f))(C(x)) &= \pi(g)(C(f(x))) = C(g(f(x))) \\ &= C((g \circ f)(x)) = \pi(g \circ f)(C(x)). \end{aligned}$$

So $\pi(g) \circ \pi(f) = \pi(g \circ f)$.

- d. For any $x \in X$,

$$\pi(\text{Id})(C(x)) = C(\text{Id}(x)) = C(x).$$

So $\pi(\text{Id})$ is the identity map of $\pi(X)$.

- e. By the previous two parts,

$$\pi(f) \circ \pi(f^{-1}) = \pi(f \circ f^{-1}) = \pi(\text{Id}) = \text{Id},$$

and similarly, $\pi(f^{-1}) \circ \pi(f) = \text{Id}$. So $\pi(f^{-1})$ is the inverse to $\pi(f)$, and so $\pi(f)$ is a bijection. ■

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

Prove that a discrete metric space is compact if and only if it is finite.

Solution. Suppose X is discrete and compact. Then $\mathcal{U} := \{\{x\} \mid x \in X\}$ is an open cover of X and therefore has a finite subcover which implies that X is finite.

If a discrete space is finite then it is compact as $\{\{x\} \mid x \in X\}$ is a finite subcover of any open cover of X . ■

Exercise 2.

Given two compact subspaces A and B of a metric space X , prove that $A \cup B$ and $A \cap B$ are compact.

Solution. Let $\mathcal{U}$ be an open cover of $A \cup B$. Then $\mathcal{U}$ is an open cover of both A and B so there exist finite subcovers $\mathcal{U}_A$ and $\mathcal{U}_B$ of A and B respectively. So $\mathcal{U}_A \cup \mathcal{U}_B$ is a finite subcover of $\mathcal{U}$.

Now $A \cap B$ is closed subspace of $A \cup B$ since A and B are both closed being compact. Therefore $A \cap B$ is a closed subspace of a compact space $A \cup B$, and therefore compact. ■

Exercise 3.

Let (X, d) be a compact metric space and suppose that $f: X \rightarrow X$ is a continuous map such that $f(x) \neq x$, for all $x \in X$. Prove that there exists $\varepsilon > 0$ such that $d(f(x), x) \geq \varepsilon$, for all $x \in X$.

Hint: Consider $g: X \rightarrow \mathbf{R}$, $g(x) = d(f(x), x)$.

Solution. The function $g: X \rightarrow \mathbf{R}$ given by $g(x) = d(f(x), x)$ is continuous being a composition of continuous functions ($g = d \circ (f \times \text{Id}) \circ \Delta$ where $\Delta(x) = (x, x)$). Since $f(x) \neq x$, for all $x \in X$, $g(X) \in (0, +\infty)$. Since X is compact, g attains its bounds on X and so in particular there exists $\varepsilon > 0$ such that $\varepsilon = \inf g(X)$ which gives the result. ■

Exercise 4.

Prove that the following are equivalent:

- I. X is compact;
- II. if $\{V_\alpha \mid \alpha \in I\}$ is a collection of closed subsets of X such that $\bigcap_{\alpha \in J} V_\alpha \neq \emptyset$ for any finite $J \subseteq I$, then $\bigcap_{\alpha \in I} V_\alpha \neq \emptyset$.

Solution. “ $\Rightarrow$ ” Suppose that X is compact and let $U_\alpha := X \setminus V_\alpha$. If $\bigcap_{\alpha \in I} V_\alpha = \emptyset$, then $X = X \setminus (\bigcap_{\alpha \in I} V_\alpha) = \bigcup_{\alpha \in I} U_\alpha$, i.e. $\{U_\alpha \mid \alpha \in I\}$ is an open cover of X . Since X is

compact there exists a finite subcover $\{U_\alpha \mid \alpha \in J\}$, for some finite subset $J \subseteq I$, i.e.

$$X = \bigcup_{\alpha \in J} U_\alpha \implies X \setminus \left(\bigcup_{\alpha \in J} U_\alpha \right) = \bigcap_{\alpha \in J} V_\alpha = \emptyset,$$

which contradicts the assumption.

“ $\Leftarrow$ ” Let $\mathcal{U} = \{U_\alpha \mid \alpha \in I\}$ be an open cover of X and $V_\alpha := X \setminus U_\alpha$. Then

$$X = \bigcup_{\alpha \in I} U_\alpha \implies X \setminus \left(\bigcup_{\alpha \in I} U_\alpha \right) = \bigcap_{\alpha \in I} V_\alpha = \emptyset.$$

So there exists a finite subset $J \subseteq I$ such that $\bigcap_{\alpha \in J} V_\alpha = \emptyset$, which implies that $X = \bigcup_{\alpha \in J} U_\alpha$. So $\{U_\alpha \mid \alpha \in J\}$ is a finite subcover of X . ■

Exercise 5.

Suppose that X is a compact metric space and $\{V_n \mid n \in \mathbb{N}\}$ is a collection of non-empty closed nested subsets of X , that is, $V_{n+1} \subseteq V_n$ for all $n \in \mathbb{N}$. Then $\bigcap_{n \in \mathbb{N}} V_n \neq \emptyset$.

Solution. Follows from the previous exercise. ■

*Exercise 6.

With the notation of Exercise 5, suppose that U is open in X and that $\bigcap_{n \in \mathbb{N}} V_n \subseteq U$. Prove that $V_n \subseteq U$, for some $n \in \mathbb{N}$.

Hint: Consider the sequence of sets $W_n = V_n \cap (X \setminus U)$.

Solution. The W_n are closed in X and are also nested. Therefore, $\bigcap_{n \in \mathbb{N}} W_n \neq \emptyset$ by Exercise 5, if the W_n are all non-empty. But $\bigcap_{n \in \mathbb{N}} W_n = (\bigcap_{n \in \mathbb{N}} V_n) \cap (X \setminus U)$ which implies that $\bigcap_{n \in \mathbb{N}} V_n \not\subseteq U$ contradicting our assumption. Therefore, at least one of the W_n must be empty implying that $V_n \subseteq U$. ■

*Exercise 7.

Suppose that X is a compact metric space and suppose that $f: X \rightarrow X$ is a continuous map. Let $X_0 = X$, $X_1 = f(X_0)$ and inductively define $X_{n+1} = f(X_n)$ for $n \geq 1$.

a. Show that $A = \bigcap_{n \in \mathbb{N}} X_n$ is non-empty.

Hint: Remember Exercise 5.

b. Show further that $f(A) = A$.

Hint: To show that $a \in A$ is in $f(A)$ apply Exercise 5 to the sets $V_n = f^{-1}(a) \cap X_n$.

- Solution.* a. We show that the X_n form a nested sequence of closed subsets of X . Note that $X_1 = f(X_0) = f(X) \subseteq X_0$. Suppose inductively that $X_n \subseteq X_{n-1}$ for some integer $n \geq 1$. Then $f(X_n) \subseteq f(X_{n-1})$ which says $X_{n+1} \subseteq X_n$. So by induction $X_n \subseteq X_{n-1}$ for all $n \in \mathbf{N}$. Also, since X is compact and f is continuous, $X_1 = f(X)$ is compact. Suppose that X_n is compact for some integer $n \geq 0$. Then since f is continuous, $X_{n+1} = f(X_n)$ is also compact. By induction X_n is compact for all integers $n \geq 0$. So each X_n is closed in X . Also, each X_n is non-empty by inductive construction. Now by Exercise 5, $A = \bigcap_{n \in \mathbf{N}} X_n$ is non-empty.
- b. The inclusion $f(A) \subseteq A$ is straightforward: if $a \in A$ then $a \in X_n$ for any integer $n \geq 0$, so $f(a) \in f(X_n) = X_{n+1}$ for any integer $n \geq 0$. But since $X_0 \supseteq X_1 \supseteq \dots \supseteq X_n \supseteq \dots$, we have $\bigcap_{n=0}^{\infty} X_{n+1} = \bigcap_{n=1}^{\infty} X_n = \bigcap_{n=0}^{\infty} X_n = A$, and we see that $f(a) \in A$. To prove the other inclusion we follow the hint: for any $a \in A$ let $V_n = f^{-1}(a) \cap X_n$. Since the X_n are nested so are the V_n . Then $\{a\}$ is closed in A , hence since f is continuous, $f^{-1}(a)$ is closed in X . Also each X_n is closed in X as in a. So each V_n is closed in X . Moreover, for each integer $n \geq 0$ we know $a \in X_{n+1} = f(X_n)$ so there exists $x \in X_n$ such that $f(x) = a$. This says that $V_n = f^{-1}(a) \cap X_n \neq \emptyset$. Now by Exercise 5, $\bigcap_{n=0}^{\infty} V_n$ is non-empty. Let b be a point in this set. Then $b \in V_n = f^{-1}(a) \cap X_n$ for any integer $n \geq 0$. Also, $b = f^{-1}(a)$ says that $f(b) = a$, and $b \in X_n$ for all integers $n \geq 0$ says that $b \in \bigcap_{n=0}^{\infty} X_n = A$. So $A \subseteq f(A)$. We have now proved that $f(A) = A$. ■

Exercise 8.

Which of the following subspaces of $\mathbf{R}$ and $\mathbf{R}^2$ are compact?

- $[0, 1)$;
- $[0, \infty)$;
- $\mathbf{Q} \cap [0, 1]$;
- $\{(x, y) \in \mathbf{R}^2 \mid x^2 + y^2 = 1\}$;
- $\{(x, y) \in \mathbf{R}^2 \mid |x| + |y| \leq 1\}$;
- $\{(x, y) \in \mathbf{R}^2 \mid x^2 + y^2 < 1\}$;
- $\{(x, y) \in \mathbf{R}^2 \mid x \geq 1, 0 \leq y \leq 1/x\}$.

Solution. d and e are compact. The others are not. ■

Exercise 9.

In the proof of the Heine-Borel Theorem we used the following fact: if $A \subseteq \mathbf{R}^n$ is bounded, then there exists $a, b \in \mathbf{R}$ such that $A \subseteq [a, b]^n$. Prove this.

Solution. If A is bounded, then there exists $r > 0$ such that $A \subseteq \overline{B}_r(x)$, for some $x \in A$. Taking $a = \min\{x_i - r \mid i = 1, 2, \dots, n\}$ and $b = \max\{x_i + r \mid i = 1, 2, \dots, n\}$ we see that $\overline{B}_r(x) \subseteq [a, b]^n$. ■

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

Suppose that $f: (0, 1) \rightarrow \mathbf{R}$ is continuous, monotonic and bounded on $(0, 1)$. Prove that f is uniformly continuous.

Solution. Suppose that f is increasing. Then we can prove

$$\lim_{x \rightarrow 0^+} f(x) = \inf_{x \in (0,1)} f(x) \quad \text{and} \quad \lim_{x \rightarrow 1^-} f(x) = \sup_{x \in (0,1)} f(x).$$

We can now define $\bar{f}: [0, 1] \rightarrow \mathbf{R}$ by

$$\bar{f}(x) := \begin{cases} \inf f(x), & x = 0; \\ f(x), & x \in (0, 1); \\ \sup f(x), & x = 1. \end{cases}$$

Then $\bar{f}$ is a continuous function from a compact domain and so is uniformly continuous. Therefore, $f = \bar{f}|_{(0,1)}$ is uniformly continuous. ■

Exercise 2.

Show that two Lipschitz equivalent metrics are uniformly equivalent.

Solution. Suppose that d_1 and d_2 are Lipschitz equivalent metrics on X , i.e. there exists $h, k \in (0, \infty)$ such that

$$hd_2(x, y) \leq d_1(x, y) \leq kd_2(x, y), \quad \forall x, y \in X.$$

Then we need to show that for every $\varepsilon > 0$, there exists $\delta_1, \delta_2 > 0$ such that

$$d_1(x, y) < \delta_1 \implies d_2(x, y) < \varepsilon \quad \text{and} \quad d_2(x, y) < \delta_2 \implies d_1(x, y) < \varepsilon.$$

We can take $\delta_1 = h\varepsilon$ and $\delta_2 = \varepsilon/k$. ■

Exercise 3.

Show that for any metric space (X, d) , the metrics d and $d'(x, y) := \frac{d(x, y)}{1+d(x, y)}$ are uniformly equivalent but may not be Lipschitz equivalent.

Solution. Note that $0 \leq d'(x, y) < 1$, for all $x, y \in X$. We need to show that for all $\varepsilon > 0$, there exists $\delta, \delta' > 0$ such that

$$d(x, y) < \delta \implies d'(x, y) < \varepsilon \quad \text{and} \quad d'(x, y) < \delta' \implies d(x, y) < \varepsilon.$$

We can take $\delta = \varepsilon$ and $\delta' = \varepsilon/(\varepsilon + 1)$ since

$$d'(x, y) \leq d(x, y) < \delta \quad \text{and} \quad d(x, y) = \frac{d'(x, y)}{1 - d'(x, y)} < \frac{\delta'}{1 - \delta'}.$$

On $\mathbf{R}$, if d and d' were Lipschitz equivalent then there would exist $k > 0$ such that

$$d(x, y) = |x - y| \leq kd'(x, y) < k, \quad \forall x, y \in \mathbf{R},$$

which is obviously false. ■

Exercise 4.

Show that the metric on $(0, 1)$ given by $d(x, y) := |x^{-1} - y^{-1}|$ is topologically equivalent but not uniformly equivalent to the Euclidean metric on $(0, 1)$.

Solution. Let d_2 denote the Euclidean metric on $(0, 1)$. Since $d_2(x, y) = |x - y| \leq d(x, y)$, if a subset $U \subseteq (0, 1)$ is d_2 -open, then it is d -open. Given $a \in (0, 1)$ and $\varepsilon > 0$, let $\delta = \min\{\frac{1}{2}a^2\varepsilon, a/2\}$. If $|x - a| < \delta$, then

$$d(x, a) = \frac{|x - a|}{|xa|} < \frac{2|x - a|}{a^2} < \varepsilon.$$

Therefore, the identity map is (d_2, d) -continuous, i.e. if U is d -open, then it is d_2 -open. So d and d_2 are topologically equivalent.

But the identity map is not uniformly (d_2, d) -continuous, since taking $\varepsilon = 1$ it is not possible to find a $\delta > 0$ such that $|x - y| < \delta$ implies $d(x, y) < 1$. ■

Exercise 5.

Prove that if $A \subseteq \mathbf{R}$ is not closed, then there exists a continuous function $f: A \rightarrow \mathbf{R}$ that is not uniformly continuous.

Solution. If A is not closed then there exists a point $x \in \overline{A} \setminus A$. Define $f: A \rightarrow \mathbf{R}$ by $f(x) = 1/(x - a)$. Then this is a continuous function that is not uniformly continuous. ■

Exercise 6.

Prove that any Cauchy sequence in a metric space is bounded.

Solution. Let (x_n) be a Cauchy sequence in a metric space (X, d) and take $\varepsilon = 1$. Then there exists $N \in \mathbf{N}$ such that $m, n \geq N \implies d(x_m, x_n) < 1$. Let

$$M := \max\{d(x_i, x_N) \mid i \in \{1, \dots, N-1\}\}.$$

Then $d(x_n, x_N) \leq \max\{1, M\}$, for all $n \in \mathbf{N}$, and so (x_n) is bounded. ■

Exercise 7.

Give an example of a metric space that is totally bounded but not compact.

Solution. The subspace $(0, 1) \subseteq \mathbf{R}$ is totally bounded (in general, bounded and totally bounded are the same in Euclidean space) but it is not compact. ■

Exercise 8.

Find appropriate sequences that prove the following subspaces of Euclidean space are not sequentially compact:

- a. $(-6, 9) \subseteq \mathbf{R}$;
- b. $\mathbf{Z}^2 \subseteq \mathbf{R}^2$;
- c. $\{x \in \mathbf{R}^3 \mid |x_1| > 2\} \subseteq \mathbf{R}^3$;
- d. $\{x \in \mathbf{R}^4 \mid x_1 x_2 x_3 x_4 < 0\} \subseteq \mathbf{R}^4$.

Solution. In each subspace we need to find an infinite sequence with no convergent subsequences (in that subspace!) The following are suitable examples (amongst others):

- a. $(1/n - 6)$;
- b. (n, n) ;
- c. $(3, 0, n)$;
- d. $(n, -1, 1, 10)$.

***Exercise 9.**

Suppose that X is a sequentially compact metric space and that $\mathcal{U} = \{U_1, U_2, \dots, U_n\}$ is an open cover of X . Put $C_i := X \setminus U_i$. Show that if $U_i = X$, for some $i \in \{1, \dots, n\}$, then any $\varepsilon > 0$ is a Lebesgue number for $\mathcal{U}$. From now on, assume that no U_i is X .

- a. Defining $d(x, C_i) = \inf\{d(x, c) \mid c \in C_i\}$, prove that $x \mapsto d(x, C_i)$ is a continuous real-valued function on X all of whose values are non-negative.
- b. Defining $f(x) = \frac{1}{n} \sum_{i=1}^n d(x, C_i)$, show that $f: X \rightarrow \mathbf{R}$ is continuous with a positive value for each $x \in X$.
- c. Deduce that there exists $\varepsilon > 0$ such that $f(x) \geq \varepsilon$, for all $x \in X$.
- d. For any $x \in X$, show that $f(x) \leq \max\{d(x, C_i) \mid i \in \{1, \dots, n\}\}$.
- e. For any $x \in X$, show that $B_\varepsilon(x) \subseteq U_{k(x)}$, where $k(x) \in \{1, \dots, n\}$ is such that $d(x, C_{k(x)}) = \max\{d(x, C_i) \mid i \in \{1, \dots, n\}\}$. Deduce that ε is a Lebesgue number for $\mathcal{U}$.

Solution. We can just take $U(x) = U_i$, for all $x \in X$, to show that any $\varepsilon > 0$ is a Lebesgue number if $U_i = X$.

- a. Since C_i is closed in X , it is compact, and for every $x \in X$, there is a $c_x \in C_i$ such that $d(x, C_i) = d(x, c_x)$. Therefore, $d(x, c_x) \leq d(x, c)$, for every $c \in C_i$. Now let $x \in X$ and $\varepsilon > 0$. Take $\delta = \varepsilon$ and suppose $d(x, y) < \delta$. Suppose, without loss of generality, that $d(x, C_i) \geq d(y, C_i)$. Then

$$|d(x, C_i) - d(y, C_i)| = d(x, c_x) - d(y, c_y) \leq d(x, c_y) - d(y, c_y) \leq d(x, y) < \delta = \varepsilon.$$

So the function is continuous. Since $d(x, C_i)$ is the infimum of a set of non-negative numbers it is also non-negative.

- b. By the composition of continuous functions, f is continuous. For $x \in X$, x cannot be in every C_i since otherwise x would not be in any U_i and $\mathcal{U}$ would not be an open cover for X . Therefore, there exists $k \in \{1, \dots, n\}$ such that $x \notin C_k$ and $d(x, C_k) > 0$. Therefore, $f(x) > 0$, for all $x \in X$.
- c. Since f is a continuous real-valued function with compact domain it attains its bounds, in particular, its infimum. As it is positive everywhere, this implies that there exists $\varepsilon > 0$ such that $f(x) \geq \varepsilon$ for all $x \in X$.
- d. We have,

$$\begin{aligned} f(x) &= \frac{1}{n} \sum_{i=1}^n d(x, C_i) \leq \frac{1}{n} (n \cdot \max\{d(x, C_i) \mid i \in \{1, \dots, n\}\}) \\ &= \max\{d(x, C_i) \mid i \in \{1, \dots, n\}\}. \end{aligned}$$

- e. Suppose that $y \in B_\varepsilon(x)$, that is: $d(x, y) < \varepsilon$. We need to show that $y \in U_{k(x)} = X \setminus C_{k(x)}$. We know that $0 < \varepsilon \leq f(x) \leq d(x, C_{k(x)})$. If $y \in C_{k(x)}$, then $d(x, C_{k(x)}) < d(x, y) < \varepsilon$ providing a contradiction. ■

*Exercise 10.

Suppose that X is a compact metric space and $V_1 \supseteq V_2 \supseteq \dots$ is a nested sequence of closed subsets of X . Prove that

$$\text{diam} \left(\bigcap_{n=1}^{\infty} V_n \right) = \inf\{\text{diam } V_n \mid n \in \mathbf{N}\}.$$

Solution. If, say, V_{n_0} is empty, then $\bigcap_{n=1}^{\infty} V_n = \emptyset$ and its diameter is 0 by definition. Likewise in this case $\text{diam } V_{n_0} = 0$ so $\inf\{\text{diam } V_n \mid n \in \mathbf{N}\} = 0$ also. Suppose now that all the V_n are non-empty. (We already know that their intersection is then non-empty.) Now $\bigcap_{n=1}^{\infty} V_n \subseteq V_m$ for any $m \in \mathbf{N}$ so $\text{diam} \left(\bigcap_{n=1}^{\infty} V_n \right) \leq \text{diam } V_m$. Hence $\text{diam} \left(\bigcap_{n=1}^{\infty} V_n \right) \leq \inf\{\text{diam } V_m \mid m \in \mathbf{N}\} = m_0$ say.

Conversely m_0 is a lower bound for the diameters of the V_n so for any $\varepsilon > 0$ and any $n \in \mathbf{N}$ we know that $\text{diam } V_n > m_0 - \varepsilon$. Hence there exist points $x_n, y_n \in V_n$ such

that $d(x_m, x_n) > m_0 - \varepsilon$. Since X is sequentially compact (x_n) has a subsequence (x_{n_r}) converging to a point $x \in X$, and then (y_{n_r}) has a subsequence $(y_{n_{r_s}})$ converging to a point $y \in X$. Since $(x_{n_{r_s}})$ is a subsequence of (x_{n_r}) it too converges to x . Also, by continuity of the metric, $d(x_{n_{r_s}}, y_{n_{r_s}}) \rightarrow d(x, y)$ as $s \rightarrow \infty$. Hence $d(x, y) \geq m_0 - \varepsilon$. Also, $x, y \in V_n$ for each $n \in \mathbf{N}$ since V_n is closed in X . Since this is true for all $n \in \mathbf{N}$ we have $x, y \in \bigcap_{n=1}^{\infty} V_n$. Hence $\text{diam} \bigcap_{n=1}^{\infty} V_n \geq m_0 - \varepsilon$. But this is true for any $\varepsilon > 0$, so $\text{diam} \bigcap_{n=1}^{\infty} V_n \geq m_0$. The above taken together proves the result. ■

Exercises marked with a * are slightly harder bonus questions.

Exercise 1.

Show that in any metric space

- The union of two complete spaces is complete.
- The intersection (provided it is non-empty) of two complete spaces is complete.

Hint: Use Lemma 1.5 in the notes from this week for part (a).

Add ref

Solution. a. Let X and Y be complete metric spaces and (a_n) be a Cauchy sequence in $X \cup Y$. Then (a_n) has infinitely many terms in either X or Y (as an infinite sequence). Suppose (a_n) has infinitely many terms in X , then these terms can be considered as a subsequence of (a_n) in X which will converge to a point in X by completeness. Therefore, (a_n) converges in $X \cup Y$ by Lemma 1.5 in the notes. Add ref

- The intersection $X \cap Y$ is a closed subspace of the complete space X , and is therefore complete. ■

Exercise 2.

Prove that any discrete metric space is complete.

Solution. Let (x_n) be a Cauchy sequence in a discrete metric space X . Taking $\varepsilon = 1$ in the definition of a Cauchy sequence proves that (x_n) is eventually constant and therefore converges. ■

Exercise 3.

Which of the following are complete:

- $\{\frac{1}{n} \mid n \in \mathbf{N}\} \cup \{0\}$;
- $\mathbf{Q} \cap [0, 1]$;
- $\{(x, y) \in \mathbf{R}^2 \mid x > 0, y \geq \frac{1}{x}\}$.

Solution. a. Complete (as a closed subspace of $\mathbf{R}$);

- Not complete;

c. Complete (as a closed subspace of $\mathbf{R}^2$). ■

Exercise 4.

Give an example of topologically equivalent metrics d and d' on a set X and a sequence (x_n) in X that is Cauchy in (X, d) but not in (X, d') .

Hint: On a previous exercise sheet we had an example of topologically equivalent metrics that were not uniformly equivalent.

Solution. On $(0, 1)$, take the standard metric d_2 and $d(x, y) = |x^{-1} - y^{-1}|$. Then the sequence $(1/n)$ is d_2 -Cauchy but is not d -Cauchy. ■

***Exercise 5.**

Which of the following metrics on $\mathbf{R}$ give complete metric spaces?

- a. $d(x, y) = |x^3 - y^3|$;
- b. $d(x, y) = |e^x - e^y|$;
- c. $d(x, y) = |\tan^{-1}(x) - \tan^{-1}(y)|$.

Solution. Let us label these metrics d_a, d_b, d_c .

- a. Then $(\mathbf{R}, d_a)$ is complete. For let (x_n) be a d_a -Cauchy sequence. Then (x_n^3) is a Cauchy sequence in the ordinary sense, so it converges say to $y \in \mathbf{R}$. Let $x = y^{1/3}$. Then (x_n) converges to x in the metric d_a . This holds since

$$d_a(x_n, x) = |x_n^3 - x^3| = |x_n^3 - y| \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

- b. This is not complete. Consider the sequence $(-n)$. Then $e^{-n} \rightarrow 0$ as $n \rightarrow \infty$, so (e^{-n}) is Cauchy in the usual sense, so $(-n)$ is d_b -Cauchy. But suppose that $(-n)$ converged to a real number x_0 in $(\mathbf{R}, d_b)$. Then $d_b(-n, x_0) \rightarrow 0$ as $n \rightarrow \infty$, which says that (e^{-n}) converges to e^{x_0} in the usual metric on $\mathbf{R}$. But we know that (e^{-n}) converges to 0 in the usual metric, so would have $e^{x_0} = 0$. But there is no such real number x_0 , so $(-n)$ cannot converge in $(\mathbf{R}, d_b)$. Hence $(\mathbf{R}, d_b)$ is not complete.
- c. For similar reasons this is not complete either. Consider the sequence (n) in $\mathbf{R}$. The sequence $(\tan^{-1}(n))$ converges to $\pi/2$ in the usual sense as $n \rightarrow \infty$. So $(\tan^{-1}(n))$ is Cauchy in the usual sense which says that (n) is d_c -Cauchy. But suppose that (n) converges to some $x_0 \in \mathbf{R}$ in the metric space $(\mathbf{R}, d_c)$. Then $(\tan^{-1}(n))$ converges to $\tan^{-1}(x_0)$ in the usual sense. This gives $\tan^{-1}(x_0) = \pi/2$. But there is no such real number x_0 . So (n) does not converge in $(\mathbf{R}, d_c)$ and this space is not complete. ■

Exercise 6.

Show that the uniformly continuous image of a complete metric space is not necessarily complete.

Solution. Define $f: [1, +\infty) \rightarrow (0, 1]$ by $f(x) = 1/x$. ■

***Exercise 7.**

Suppose that $f: X \rightarrow Y$ is a bijection between metric spaces such that f is uniformly continuous and f^{-1} is continuous.

- Prove that if (x_n) is a Cauchy sequence in X , then $f(x_n)$ is a Cauchy sequence in Y .
- Prove that if y_n is a convergent sequence in Y then $(f^{-1}(y_n))$ is a convergent sequence in X .
- Deduce that if Y is complete, then X is complete.
- Deduce that completeness is an invariant of uniform equivalence.

Solution. a. Let $\varepsilon > 0$. Then, by the uniform continuity of f , there is a $\delta > 0$ such that $d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon$. For this δ , there is an $N \in \mathbb{N}$, such that $m, n \geq N \implies d_X(x_m, x_n) < \delta$ since (x_n) is Cauchy. Therefore, $m, n \geq N \implies d_Y(f(x_m), f(x_n)) < \varepsilon$.

b. Follows from the (sequential) continuity of f^{-1} .

c. Let (x_n) be a Cauchy sequence in X . Then $f(x_n)$ is a Cauchy sequence in Y by a which implies that $f(x_n)$ converges in Y since we are assuming Y is complete. Therefore, by b, $(f^{-1}(f(x_n))) = (x_n)$ converges in X and so X is complete.

d. If we suppose that f is now a uniform equivalence we immediately obtain that if Y is complete, then X is complete. By considering f^{-1} we also get the converse. ■

Exercise 8.

Let $f: (0, 1/4) \rightarrow (0, 1/4)$ be given by $f(x) = x^2$. Show that f is a contraction without a fixed point.

Solution. Firstly, $f(0, 1/4) \subseteq (0, 1/4)$. Also, f is a contraction since

$$|f(x) - f(y)| = |x^2 - y^2| = (x + y) \cdot |x - y| < \frac{1}{2}|x - y|.$$

It has no fixed point since $x^2 = x \implies x = 1 \notin (0, 1/4)$. ■

Exercise 9.

Let $X = \{x, y, z\}$ and let d be the discrete metric on X . Define

$$d'(x, y) = 2 = d'(x, z), \quad d'(y, z) = 1, \quad \text{and} \quad d'(x, x) = d'(y, y) = d'(z, z) = 0.$$

Show that d' is a metric on X that is Lipschitz equivalent to d . Define $f: X \rightarrow X$ by

$$f(x) = y, \quad f(y) = z \quad \text{and} \quad f(z) = z.$$

Show that f is a d' -contraction but not a d -contraction.

Solution. It is easy to show that d' is a metric on X . To show that it is Lipschitz equivalent to d note that

$$\frac{1}{2}d'(a, b) \leq d(a, b) \leq d'(a, b) \quad \forall a, b \in X.$$

By putting $K = 1/2$ we can see that f is a d' -contraction by checking the definition. To show that f is not a d -contraction observe that

$$d(f(x), f(y)) = d(y, z) = 1 \leq Kd(x, y) = K \implies K \geq 1.$$

Since K has to be less than 1, f is not a d -contraction. ■

Exercise 10.

Let $f: X \rightarrow X$ be a map of a compact metric space X such that

$$d(f(x), f(y)) < d(x, y)$$

for any distinct points $x, y \in X$. Prove that f has a unique fixed point.

Hint: Show that $\inf\{d(x, f(x)) \mid x \in X\}$ is attained, and get a contradiction unless this inf is zero.

Solution. Consider the composition

$$d \circ (1 \times f) \circ \Delta: X \rightarrow X \times X \rightarrow X \times X \rightarrow \mathbf{R}.$$

Each map here is continuous so the composition, F say, is continuous. Since X is compact, F attains its lower bound l , say, on X . Suppose that $l > 0$ and that it is attained at x_0 , so $d(x_0, f(x_0)) = l$. Then x_0 and $f(x_0)$ are distinct, so by assumption $l \leq F(f(x_0)) = d(f(x_0), f(f(x_0))) < d(x_0, f(x_0)) = l$. This contradiction shows that l must be 0. Since l is attained $d(p, f(p)) = 0$ for some $p \in X$, so $f(p) = p$ and we have proved the existence of a fixed point.

Uniqueness: for distinct fixed points p, q we would have

$$d(p, q) = d(f(p), f(q)) < d(p, q),$$

a contradiction. So p is the unique fixed point. ■

Exercise 1.

Suppose that $A \subseteq \mathbf{R}$ is of the form $A = \bigcup_{i=1}^r A_i$ and that a sequence of functions $f_n: A \rightarrow \mathbf{R}$ is uniformly convergent on A_i for each $i = 1, 2, \dots, r$. Show that (f_n) is uniformly convergent on A .

Solution. Let f be the pointwise limit of (f_n) on A , which certainly exists since each $x \in A$ belongs to some A_i , so $(f_n(x))$ converges. Let $\varepsilon > 0$. For each $i \in \{1, \dots, r\}$ the sequence (f_n) converges uniformly on A_i , so there exists $N_i \in \mathbf{N}$ such that $|f_n(x) - f(x)| < \varepsilon$ whenever $n \geq N_i$ and $x \in A_i$. Let $N = \max\{N_1, N_2, \dots, N_r\}$, let x be any point in A and let $n \geq N$. Then $x \in A_i$ for some $i \in \{1, \dots, r\}$ and $n \geq N \geq N_i$, so $|f_n(x) - f(x)| < \varepsilon$. Thus (f_n) converges uniformly on A . ■

Exercise 2.

Show that if $f_n \rightarrow f$ uniformly on A and if each f_n is bounded on A then

- f is bounded on A ;
- there is a uniform bound for the f_n , that is, there exists $K \in \mathbf{R}$ such that

$$|f_n(x)| \leq K, \quad \forall x \in A, \forall n \in \mathbf{N}.$$

Solution. a. There exists $N \in \mathbf{N}$ such that $n \geq N \implies |f_n(x) - f(x)| < 1$, for all $x \in A$. Let $|f_N(x)| \leq L$, for all $x \in A$. Then

$$|f(x)| < 1 + |f_N(x)| \leq 1 + L, \quad \forall x \in A.$$

b. Using the same idea as in a, we see that

$$n \geq N \implies |f_n(x)| < 1 + |f(x)| \leq 2 + L, \quad \forall x \in A.$$

For each $m < N$, we know that there exists L_m such that $|f_m(x)| \leq L_m$, for all $x \in A$. So we can take $K := \max\{2 + L, L_m \mid m = 1, 2, \dots, N - 1\}$. ■

Exercise 3.

Construct a sequence of functions $(f_n: \mathbf{R} \rightarrow \mathbf{R})$ none of which are continuous at $x = 0$, but such that (f_n) converges uniformly on $\mathbf{R}$ to a continuous function.

Solution. Take

$$f_n(x) = \begin{cases} \frac{1}{n}, & x \in \mathbf{Q}; \\ 0, & x \notin \mathbf{Q}. \end{cases}$$

■

Exercise 4.

Let $y(t) = e^t$ with $t_0 = 0$. Then calculate the first few Picard iterates $\varphi_n(t)$ of the corresponding IVP and check that it is converging to the well-known series expansion of e .

Solution. We have $y_0 = y(t_0) = 1$ and so $\varphi_0(t) = 1$. Then

$$\begin{aligned}\varphi_1(t) &= 1 + \int_0^t ds = 1 + t, & \varphi_2(t) &= 1 + \int_0^t (1 + s) ds = 1 + t + \frac{t^2}{2} \\ \text{and } \varphi_3(t) &= 1 + \int_0^t \left(1 + s + \frac{s^2}{2}\right) ds = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}.\end{aligned}$$

We know that $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$. ■